SCORE: A Framework for Synthesizing Complete, Owned, Responsive Environments

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Abstract

Games, films and robot training need 3D environments that are complete, editable and owned by their makers; world models and one-shot generators give frames no engine opens, or one fused scene without separate objects or clear rights. Studios build environments in stages, from concept art to review. SCORE keeps these stages and automates them with open vision foundation models, procedural tools and coding agents, while the creator designs and reviews. The creator locks an environment concept and picks each place's concept; the pick is read into a scene inventory and written into a plan, and only such a design starts paid work, bought as one batch with one estimate and a stop at its high estimate plus 10 percent. Every stage is a node of one runner: its output is keyed on its inputs and code, judged by gates that answer pass, fail or unknown, and accepted only when nothing got worse, so a fault is fixed in its own stage and only what follows it runs again. Every model allows commercial use of its output, so the creator stays the author. We describe the framework with its first environment, 26 places derived from the game 2099, as the worked example, and separate what its runs prove, what is built but has not run, and what is missing: the first run brought every place through its completion gate, but its video was not accepted, and the remake from new designs has not run. Code and paper are available at https://github.com/Babon-Innovations-b-v/score (MIT license).

1 Introduction

Interactive 3D environments are the raw material of games, film and embodied AI [1], and building one is expensive: every object is modelled, surfaced, placed, given collision, lit and given sound in one style, and the environment stays editable. Generative models promise to take over much of this, but in GDC's 2026 survey 52 percent of game workers thought generative AI was hurting the industry [2]; a tool for building environments has to leave the result in the creator's hands.

World models generate an environment as frames [3], [4] that no engine can open. 3D environment generators [5], [6] produce geometry, but mostly as one fused asset, from which separate objects are recovered after the fact [7], [8]. Both work in one shot, so a creator cannot find or change the step behind a fault. Procedural systems [9], [10] and coding agents that write scenes as programs [11], [12], [13], [14] keep the environment explicit, but rebuild a scene that already exists or draw their objects from a fixed library, and such agents judge their own geometry unreliably [15]. Open vision foundation models [16] now read shapes [17], masks [18], depth [19] and parts [20] out of pictures, and a coding agent can connect them as tools; none of them provides one style across objects that different models made, which the field names as an open problem [21], [22].

Recent agents write Blender programs and check them against a target: from one image, in stages of geometry, materials, composition and light [23]; from text or video, as worlds for physical reasoning

[24]; and with motion added [25], [26]. Benchmarks find that their procedural models often float or come apart [27], and that on dynamic scenes rebuilt from video the best model scores 0.91 on static scenes but 0.67 on dynamic ones; mesh quality does not track fidelity, and VLM judges agree with people in 89.3 percent of comparisons but near chance between close models [28]. SCORE builds static environments in explicit, gated stages and treats motion as its own checked layer, which this gap between static and dynamic scenes supports.

Studios build environments in stages, each with an output a person can check; SCORE keeps these stages and automates them. Its environments are meant to be **complete**: every object, surface, light and sound exists wherever the viewer looks. They are **owned**: every model in the framework allows commercial use of its output, so the creator stays the author, holding the full commercial rights. They are meant to be **responsive**: every object has real collision, mass and material. And they can be any kind of **environment**, from a house on a plot of ground to a system of worlds. To our knowledge, no other openly available environment generation work joins vision foundation models into engine-ready scenes that are built from reference pictures, steered by a person's choices at each step, and made only with models that allow commercial use of their output.

We contribute **the SCORE framework**, a design step the creator owns and stages as nodes of one runner that keys, judges and accepts every output over one canonical scene (Section 2); **measured gates** for every layer of a place and a completion gate over them (Section 3); and **evidence from a first environment**, which separates what its runs prove from what is built but has not run and what is missing (Section 4).

2 The framework

An environment is a table of stages; environment 1's has 149 rows, 89 of them one for each person's look, dress or take. A **node** is one stage for one place (`assembly@hangar`) or for the environment, and one runner makes, judges and accepts every node. The creator works in the creator cockpit, one signed-in web page: there he designs each place (Section 2.2), buys paid work as batches and reviews what the runner reports. From an approved design everything runs without a further yes.

2.1 Keys, gates and accepts

Keys. A node is made only when its key has no finish record. The key hashes the stage and place, a code version, the contracts of its output and inputs, its parameters, a seed derived from the environment's, the pinned model images, every environment file it reads and the accepted output of every node it reads. The code version hashes every source file the stage imports, leaving out the runner and the renting code, which change what a run costs, not what it makes. A fault fixed in one stage moves only its key and those after it. Outputs are kept by content hash, never overwritten.

Gates and accepts. Each gate answers pass, fail or unknown, with measures and a reason; missing evidence is unknown, and unknown blocks like fail. The runner accepts an output by itself when every gate passed and it is no worse than the one accepted before: no answer dropped, no measure moved the wrong way by a step a viewer could see, no gate missing. Later nodes read only accepted outputs. The creator keeps four actions, buttons in the cockpit applied on his own PC: reject an accepted output (it moves back to the one before, and every node that read it is stale), accept a worse one with a reason, waive one gate for one key, and override the runner for one paid run. When the creator accepts a worse output, that accept records what was allowed, and a later output no worse than the one taken is accepted with that accept named. None is needed for a run; a barrier

keeps agents from them (a store that refuses the creator's records from an agent's process, and a hook).

Batches. Paid work is bought as a batch. Before it starts, the runner gives one estimate and one ETA from measured unit costs (Section 3.7); the creator's yes to that estimate is the only spend decision, and the batch's stop is its own high estimate plus 10 percent. Inside a batch every paid job runs once, with no retry; its checks only sort each output into a success or a named failure, and a failure becomes a batch gap that the line carries to the completion gate instead of stopping. Spending more on a gap is a new batch and a new yes. A paid node is cleared only when every gate on its inputs passed, the month's books are known from the providers' bills, and its batch's stop has room for its forecast; everything that rents a machine or calls a paid model refuses without that clearance, and buying again more than EUR 5 of accepted outputs needs a written reason. Each node's review page is built from the store alone, and the creator reviews a place in a 3D viewer on the store's copy of its accepted assembly.

The creator steers: the ship. Figure 1 shows the creator's control on one piece of environment 1, the ship, in its first run. The creator's picks were found pictures of a shuttle, a one-deck cutaway, a cockpit and crew quarters; we do not show them. From them an agent planned the bridge alone (v1, a 4.6 by 4.6 m room of 27 pieces). The creator corrected the plan in words: a smaller cockpit, the rooms environment 1 needs, a long flat wedge rather than a rocket, a nose less pointed, and an outline measured from drawings rather than shaped from words. Each correction made a new plan, and the creator accepted the seventh. Every plan is ours and cost nothing; the two drawings the outline was traced from cost EUR 0.28 and are not shown. The ship's shape follows found artwork the creator chose to use: shuttle concept by PAUL (zana), found online.

2.2 From concept to scene

The design. Before any paid batch the creator decides each place's design: its concept and its plan. He first locks an environment concept, in a short interview with Gemini 3.1 Pro [29] in the cockpit: its style, his reference pictures, its first places with the places each may hold, and its **setting**, what everything stands on and in at the scale he sets: a plot of ground with a gravity and a sun path, one world, or a system of bodies that move one another, a real body taking its values from a cited table and an invented one given by its radius, mass, orbit and spin. Each thing in the setting takes the form its scale calls for: a body near enough to matter moves under the physics, a far source of light is a light, and what is too far to move is a backdrop. An environment plan, written by the plan writer below, places each top place's site on the setting's ground at true size and works out each site's sky at its scene time. Concepts are drawn layer by layer on a concept review page, each draw one Nano Banana Pro [30] call that the creator's click pays for, where he marks areas and asks for a new draw. Every new draw is sent a **blockout** first: a grey model built in code from its parent's plan at real sizes and rendered free from the fixed concept view, so the draw keeps its layout and proportions and adds the look. A place within another is drawn only once its holder is picked and planned, on a crop of the holder's concept and its own shell inside the holder's, so inside places fit the shell they stand in.

Inventory and plan. The creator's pick is the approval. It starts the scene inventory reading: Gemini 3.1 Pro lists every object of the picked concept as a row with its kind, count, anchor and size in metres, judged against the place's outline, and SAM 3 [18] boxes each row on the concept; a row without metres, or a disagreement between the two, is a named gap. One more Gemini 3.1 Pro call writes the plan in metres, element by element, as environment 1's hand-written plans were: outline, walls, openings with their swing, levels, stairs, roofs, terrain heights, each item's shape and

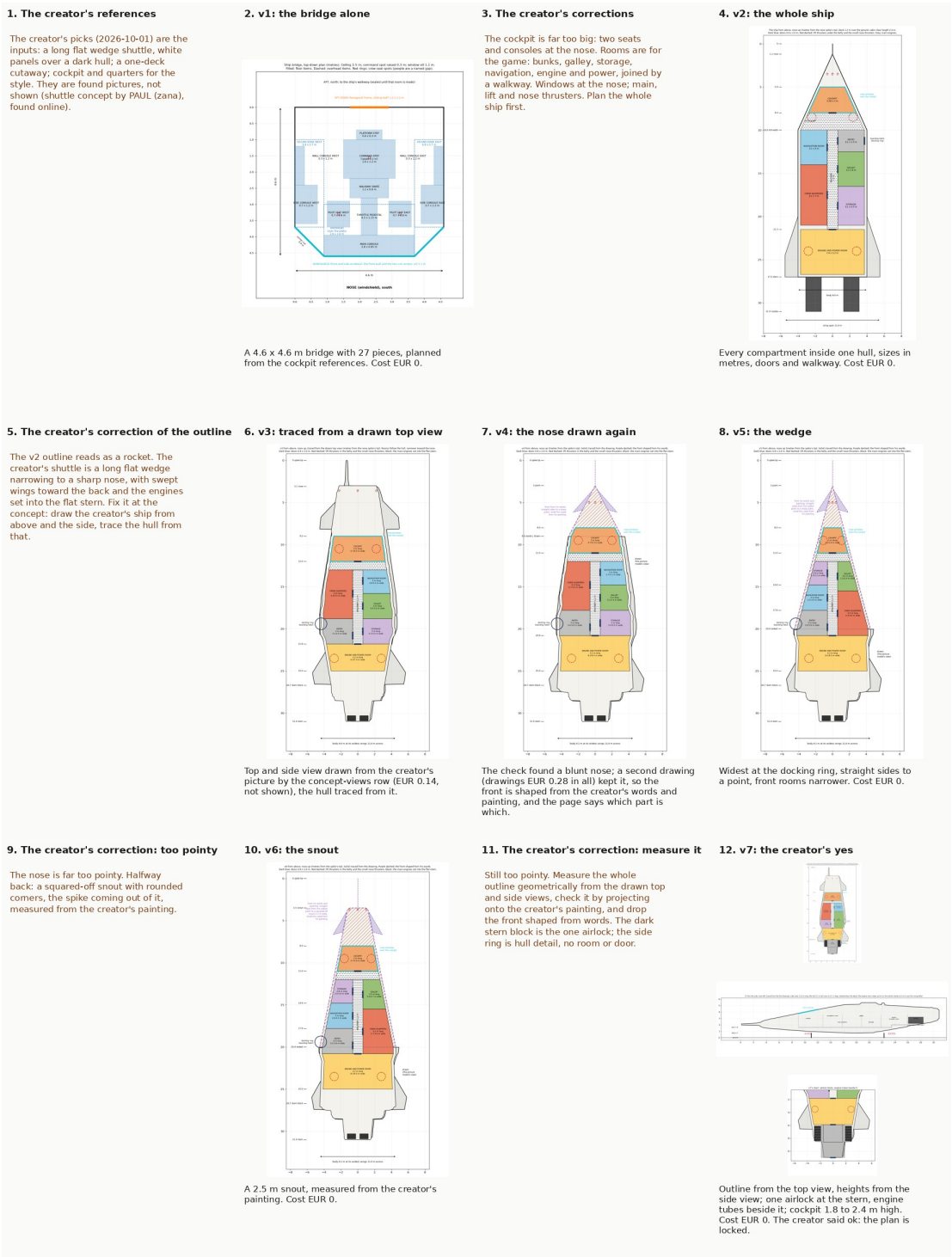


Figure 1: The creator steers the ship's plan, in twelve steps: the creator's picks, described in words (found pictures, not shown); plan v1 of the bridge alone; the creator's corrections; v2 of the whole ship; the creator's correction of the outline; v3 to v6; the creator's two corrections of the nose; v7, which the creator accepted. All plans are ours. Shuttle concept by PAUL (zana), found online.

a section. The SAM 3 boxes only order the items, since boxes from a perspective painting laid flat put objects in the wrong place. Free checks follow, each failure a gap said where it is: every row placed as often as it counts, inside the outline, no two floor items in each other, every door and standing spot reached on foot, and the inside places tiling the shell. On the plan review page the creator moves, resizes and corrects, and his edits redo what follows. Readings and plan writings run without a yes under the creator's standing allowance, their cost shown in the cockpit. Only a place with a design gets the rows of a batch.

Shapes first. The route is one loop, the same at every scale. A place starts as its big shape, the outside shell, with a concept, a plan and a rough model: the concept goes through picture to model whole, and the rough model gives the plan its proportions. The shell's plan is then split into the shapes of the places inside it (rooms, corridors, holds), each sized inside the shell. Once those shapes are fixed, every place's concept, rough model, close-ups and finishes are made in parallel, since from then on each is a matter of proportions inside a fixed shape; zooming into an inside place repeats the same steps. This is the creator's order of 19 October 2026. The runner has its two rows: `rough-model` puts a place's concept through Pixal3D whole after its plan, and `shell-split`, free, cuts an outside shell's plan into its inside places' shapes, gated on `tiled`, `reached` and `bounded`, which each inside place's plan reads. Neither has run on environment 1: no rough model has been made, and no plan of it has a split yet.

Plan and scene record. Nothing is made or placed that is not an inventory row. The plan stage is gated on `mapped` (every element a row, a door or a written drop; every door says where it leads). The scene record is written from the plan, inventory, made ground and sky, gated on `game-free` (the record cites nothing from the game the environment came from), `plan-matched` (the shell within 5 cm of the plan), `openings` (every doorway onto a floor, no view into nothing) and `seed-matched` (every old entry matched or dropped with a reason). The `game-free` gate sees cited numbers only. In environment 1's first run a coding agent wrote the inventories and plans, and in 18 of its places the data still took positions from the source game (Section 4).

The canonical scene. Every place is one OpenUSD stage [31]; each object, one per real-world object, carries its kind, transform, collision and each part's library surface. Its layers lie in a fixed order: a slot for the creator's edit layer on top, then people, sound, light, motion, effects, crops, machines, settle, ground and the generated base. The edit layer exists only as that slot, an empty layer the exporter makes once and never writes again; a test on a two-box place shows an edit there surviving a regenerated base, but the runner's assembly does not yet carry a creator's edit from one assembly to the next. Each place is assembled once, from every layer's accepted output; its `layers` gate answers unknown while a needed layer has none, and an old output never stands in.

Models, generation first. Every object goes the prop route first: Nano Banana Pro draws its close-up once from its crop of the concept, Pixal3D [17] builds it, the surface stage paints it, and a likeness gate compares its outline from eight sides with its close-up. A close-up that fails the shape check's measured half (a plain border, one object filling the frame, the row's proportions) is a batch gap, so Pixal3D is never paid for it. Only an object whose checks fail is built by a code builder a coding agent writes, reversing the earlier parts check (the creator's choice, 12 October 2026). On 13 of the hub's objects both orders end the same (Table 1, Figure 2), but generation first draws a close-up of every object. Every close-up of eight thin and flat objects failed the shape check, so code builds carry them. The hatch wheel shows the gain: its code build's outline overlap was 0.45, a fail, and its generated model's 0.60.

Table 1. How 13 hub objects, chosen to cover the kinds, end in each order. Close-ups cost EUR 8.12 and the sample's prop route EUR 13.80, about EUR 2.20 an object; no take was baked (the run stopped at the keyboard's paint gate).

Order	Prop route	Code build	Needs a new builder
Generation first	4	7	2
Code first	4	7	2

Big objects seen up close. A generated model is right in its details and lumpy in its main shapes, which shows on a big object the viewer walks up to, so such an object gets a body built in code under only the generated details its entry names (a console's screens and lamps), both baked as one model in library surfaces (the creator, 18 October 2026). A re-audit of environment 1's 479 code-built inventory rows kept 396 as code, sent 80 to the prop route and 3 to a code body with generated details, and listed 46 picture-made kinds that are big objects seen up close; eight have a body builder (a cockpit console, a station module, a power core, fuel and water tanks, a lander, a rover and a crane). A prop whose body has no builder stops before its first paid step. The trial of the bodied props has not run.

3 Stages and their gates

3.1 Surfaces: the finish route

Everything a place shows comes from the creator's library, score_lib, or is a written exception: every model, kit, finish, effect, light, sound, character, ground and algorithm (the creator's rule of 18 October 2026). Models give shape only, picture-made and code-built alike, so one style holds across them: a picture-made model keeps its geometry and its finish is replaced, never kept as the picture's own texture, its baked-in colour or a default grey. Every part takes one surface from a shared library of rule-based surfaces, written as ProcFunc functions [32] on Infinigen's shaders [9], coloured from the place's palette and worn only where wear has a cause (Appendix D).

The finish route gives each model its finishes (the creator's choice of 20 October 2026). Free renders of the raw model cover every face from as many angles as a coverage check needs, and each view is split into colour regions of the model's own texture. One Gemini 3.1 Pro answer a model, a place's models in one batch, lists its finishes, each matched first to a library surface with its settings (palette token, roughness, metal, seed), and labels each region of each view. Before anything is baked, a free check compares each region with its label's surface and makes a model whose colours lie more than ΔE 32 off a gap; the labels then vote across views into one finish a face, and a face no label reaches or whose views disagree is a gap too. Each part is painted from its library surface and baked in one paid job, and one batch recheck against the close-up sorts each model into a pass or the named gap "no library surface fits". This route replaced the finish-map route of environment 1's first run, in which Nano Banana Pro drew the close-up again with one flat colour per finish, up to three times, and SegviGen [33] split the mesh along that map; that route gave every piece its own rust and stains until the parts were painted from the library (Appendix A). The finish route has run its free views on two places; its paid steps have not run.

The bake is checked against the close-up. A finish's surface must lie within ΔE 32 of its colour there, else a nearer surface of its family is taken or a new one is proposed to the creator, and the part fails by name. From the greenhouse's water tank, which the creator found baked as three near

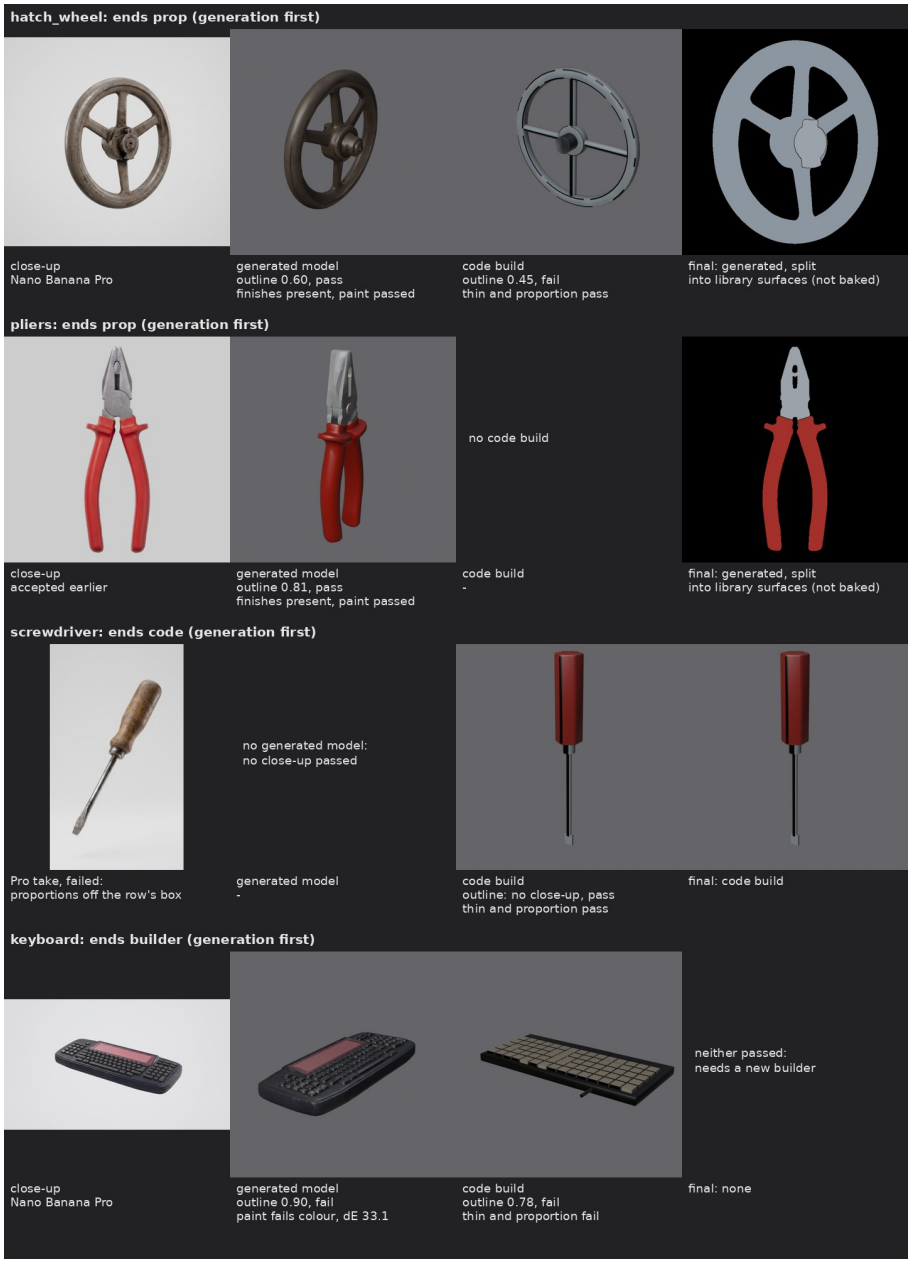


Figure 2: Generation first on four hub objects. Columns: close-up, generated model, code build, final model. The hatch wheel's generated model passes its outline check at 0.60 where its code build fails at 0.45; the pliers pass at 0.81; every close-up of the screwdriver failed, so its code build carries it; the keyboard fails both ways (paint ΔE 33.1; code build too thin and out of proportion) and needs a new builder. Generated models are split into library surfaces, not baked.

greys and tipped, came two checks: a bake fails when it darkens a finish the close-up shows lighter, or turns all surfaces into one grey; and short legs are stretched onto one level plane, a foot over 3 cm off the floor failing. The bake's finish borders follow the model's real edges, as the cut-down copy it bakes onto is shaded smooth except at folds of 60 degrees or more.

A written exception names the place, the object, what it is, why it is not from the library, who allowed it and when; only the creator allows one, and one without a reason or an owner refuses the whole file. A free library check traces every material, model, emitter, light, sound source and character of a place's accepted assembly to a library entry, by content hash or a recorded origin, and the cockpit shows what does not trace as the place's problems. It is not a gate of the completion gate, since wiring it in would move the keys of paid rows; instead the library rows at the end of every line, which hand each accepted output to the library, fail when an output is covered by no entry or written exception, so nothing reaches the library untraced. Each new entry then gets a preview clip, rendered in the batch with the films' renderer.

3.2 Resting

Loose objects are dropped onto the made ground in a physics simulation; one that would turn more than 15 degrees or drift more than 30 cm is a resting fault of the layout and keeps its laid pose. The settle stage lays where each object comes to rest as a layer (held objects snapped onto their holders, a stack onto one support, an object that would rest worse kept as laid), gated on `rested` and `resting` (no fault that appears with the layer and not without it).

3.3 Light

Lamp strengths are measured, never copied from the source game. The light layer places live lamps from the scene record, with the sky and its bodies. One paid render per place gives every lamp its own Cycles light group on the place's views; because light adds up, the look at any strengths is mixed from these passes without rendering again, and a free sweep chooses the scale that brings the place's brightness within 1.5 times its concept's, halving any lamp that turns what it lights white. Outdoors the suns are swept after the lamps. A lamp whose fixture stops 90 percent of its rays moves out of it, and one behind a clear lens stands just outside, as Cycles passes no light through a glowing lens.

3.4 Motion, effects and sound

Motion. Each moving thing (a path, a hinge, a spin, a rail, a turn, a climb) has its kind's speed range cited from real machines, gated on `clear` (no moving thing within 3 cm of a still thing or another moving thing, read every quarter second), `speeds`, `loops`, `grounded` and `covered` (every still prim of a moving kind says why). **Effects** are emitters measured against real footage (photographs of lift-offs, rover films of Martian dust devils), gated on `attached`, `footage`, `light` (the flame lights the pad at least twice as brightly as the place's lamps) and `outside` (no open-sky particle inside a room). **Sound:** every source a place names (room tones, objects, motions, each surface's steps) gets the environment's picked file or one MOSS-SoundEffect [34] take of its brief, gated on `sources`, `level`, `likeness` (CLAP [35] score at least 0.30, the tenth percentile of the environment's picks; a take under it is a gap) and `no freesound`.

3.5 People

People are made like props: a FLUX.2 klein [36] close-up in SOMA-X's bind pose, built by Pixal3D, painted by the surface line, over SOMA-X's body fitted just inside (Gauss-Newton, 24 shape numbers), whose skin weights each look point takes from its closest triangle; Kimodo's clips [37], from sentences, move it. The **shell gate** passes a look when 99 percent of its soft points lie within 5 cm of the body, its worst point recorded. Hard parts ride one joint, belts are cloth, each clip gets its own verdict (tucks, hard parts stretched, cloth or body through cloth) and casts use only passing clips; a far-only look is judged on its far body (Appendix B). **Casts:** SAM 3 finds the concept's figures, a judge tells people from pictures, reflections and empty suits, and casts are gated on `counted`, `standing`, `clear`, `made` and `drapes`; no crowd copy fills a tenth of a view's height.

3.6 Completion

A place is complete when its gate passes on its final assembly: every row made or dropped with a reason, nothing unmade, no placeholder or resting fault, the creator's picks shown, every layer and shown place present, nothing cited from the source game. **Accepted gaps.** A fault the creator decides to ship with is written down as an accepted gap: who decided, when, why, the objects and their measured faults. It moves only those objects' faults from the gate where they arise to the completion gate, which fails each gap by name with the numbers: a place whose accepted gaps are its only fail is complete apart from accepted gaps, every gap listed with who decided it; any other fault leaves it not complete, and the gate is never loosened. A batch gap moves the same way, the creator's yes to the batch being its decision, and is listed with the batch run that made it. A hook stops an agent claiming a place is done while its gate fails: agents judging their own scenes agree with a measured geometry score only 45.8 percent of the time [15].

3.7 Prices and ledgers

Machines and Google calls write ledger lines with their node's clearance; the month's spend is the higher of ledgers and bill, which caught misses: EUR 1,554 billed against EUR 1,387 in the cloud ledger on 12 October 2026, and EUR 549 billed by Google for a day the earlier ledger put at EUR 30. Over 13.8 hours of environment 1's first run the ledgers hold EUR 117.45 of machines and EUR 34.19 of Google calls, most for painting (EUR 37.61), the finish maps (EUR 26.59), the people's bodies (EUR 16.09) and the light passes (EUR 10.17).

Prices. Every paid stage is priced from a unit cost measured on the ledgers (a close-up EUR 0.07 from 135 calls; a Pixal3D model EUR 0.11 likely and 0.20 high from 52 jobs; a preview clip EUR 0.16 from one run of six), times the units its forecast rule counts, times the bill's share over the ledger. A stage with no measured run has no price: its estimate is shown apart as a guess and never added to a price. The finish route's list and recheck are priced from the earlier route's 275 list calls at the batch tier's half price, not yet from the route itself. The batch estimate and the runner use the same unit prices, and on 21 October they agreed to the cent per stage and per model on every designed place whose line could start. Before, the runner priced a run from a flat figure in its table, which overstated work it could carry over: the four runs of 15 October were forecast at up to EUR 72.57, 33, 70.50 and 24 and cost 10.13, 10.14, 0.67 and 0.60, while measured unit costs held (22 close-ups for the ship estimated at EUR 3.10 to 6.20, actual 3.50; seven box checks 1.75, actual 2.19; its eight shell kits 1.20, actual 0.60). One ship run spent EUR 13.58 before it stopped on missing free inputs; since then every input is checked free before a paid run.

4 Evidence

Environment 1 is the worked example: 26 places, from a city square and a flat to a Moon base, a ship and Mars, derived from the game 2099. Its first run ended on 18 October; its remake from new designs has not run. Table 3 sorts the framework's parts into what these runs prove, what is built but has not run, and what is missing.

The first run. Each place was filmed once its completion gate passed or failed only on its accepted gaps: a 2 s title card, then one walk of 4 s (96 frames at 24 fps) on a path the planner keeps clear of walls, objects and people, with the place's own sound. The film gate measures each film (its size, a sound track that is not silent, no black or frozen stretch after the title card, its length and the clearance) and the runner accepts it; no person reviewed the films before the cut, and the coding agent that made the cut checked only its order, frames and sound. On 18 October the cut held all 27 films, 26 places and the ship's hull, 3 min 9 s (Figure 3). The final completion gate, run again on every place on 17 and 18 October, left all 26 places complete apart from accepted gaps (2 to 19 each) and none complete; many of those gaps were decided by the overseer, the coding-agent session that guards the run under the creator's rules, and wait for the creator's confirmation. The creator watched the video on 18 October and did not accept it, naming five faults: the ship looks poor, weakest where his own direction stopped; the films cut too fast, about 3.5 s a place; no people appear; the sound is only ambient, with no voices or event sounds; and moving things do not move convincingly. On 19 October the disk holding the store, both spend ledgers and the work folder was lost; the code, the cloud bucket and the library's entry records survived, and the store is rebuilt only by today's keys and gates, so most of the run's outputs must be made again.

Layouts from the source game. In the first run the framework's own layout, made from the plan and the concept, was proven only on the ship's 8 places and the 2 door kits. The other 18 places took their layouts from the source game: 507 entries in their data cite the game, 338 of them positions, and 7 kits are laid in the game's frame, with 6,432 piece positions. The check sees cited numbers only, so an uncited copy passes: the launch's accepted scene record still lit the rocket at the game's position. Each of the 18 places shipped this as an accepted gap of the completion gate's game check. The remake's plans are written from each place's own concept and setting instead.

People. Scored on what is drawn, no clip of any look is clean (Table 2). Takes in a wider A-pose crossed less at the bind and more in motion (10,610 visible crossings on the player's clips against 4,078), with no clip clean in either pose. People therefore shipped as accepted gaps in the first run: the characters layer lays the people whose bodies stand accepted, and the completion gate fails every missing one by name.

Table 2. People, 13 October (experimental, Section 3.5): usable clips of 19 per look by the gate as built, on drawn surfaces only, and as one piece on what cameras see. Faults and fixes: Appendix B.

Look	Body gates	Gate	Visible	Cameras, one piece
Player (work)	pass	5	0	0
Bram (work)	pass	5	0	not run
Sefa (work)	pass	0	0	not run
Nev (work)	pass	0	0	not run
Kit, average man (olive)	fail: triangles	0	not run	not run
Player, wider pose (work)	fail: triangles	0	not run	0
Kit, average man, wider pose (olive)	fail: shell	0	not run	0

The design step, 20 and 21 October. For the remake the creator drew 95 concepts, EUR 14.28



Figure 3: The environment 1 video (demo K9) of the first run, 18 October: the middle frame of each of its 27 films' walks, in story order. Every frame is the runner's accepted film of a place that was complete apart from accepted gaps; nothing was retouched or picked by hand. The creator did not accept this video.

together, and picked 18 of them. Their readings (41, one refused) cost EUR 2.56: the latest of each place holds 560 rows, 541 of them with metres on three sides, and 893 gaps where the reading and SAM 3 disagree or a row has no metres. The plan writer wrote all 18 plans without a refusal, EUR 0.13 to 0.74 and 62 to 325 s each, EUR 5.09 together, and the plan row accepted 17 of them (the base, an outside shell with no plan record of its own, has none). The environment plan was written twice, EUR 0.20 each. Eight draws, one for each of the ship's rooms, were drawn on a blockout; none is picked yet.

The remake. On 21 October the estimate for the batch of the 18 designed places stood at EUR 186.50 likely and 271.89 high for 456 models, with 45 gaps predicted from the hub's measured failures priced apart and not bought, and an ETA of at least 247 minutes with nine layers still untimed. The runner's own plan for every line priced 158 of its 231 paid rows from measured units, EUR 208.87 likely and 314.78 high; the other 73 rows (EUR 194.23 of estimates, among them the films, rough models, people takes, light passes and surface tiles) have no measured run and so no price. Two benchmark batches measured stages before it: six preview clips for EUR 0.96, five right and one effect blank and refused by its check, and eleven sound takes for EUR 0.05, two of them gaps by their likeness check. Nothing of the remake has run.

Table 3. What environment 1's runs prove, what is built but has not run, and what is missing, on 21 October.

Part	Status	Record
Runner: keys, gates, automatic accepts, stale nodes	proven: first run, every place	Section 2; K9
Clearance, ledgers, books from the bills	proven: caught ledger misses	Section 3.7
Surfaces (finish-map route), resting, light, motion, effects, sound, completion gate, films	proven on the first run; video not accepted	Section 4; K9
Environment concept, concept review, inventory reading, plan writing and checks	ran at 18 places	design step
Measured prices, estimate and runner in agreement	priced; no batch spent	Section 3.7
Preview clips, one take a sound	one benchmark each	the remake
One-shot batch, its stop, batch gaps	built, tested on fixtures; no batch bought	Section 2.1
Finish route	free views at 2 places; paid steps not run	Section 3.1
Shapes first: rough model, shell split	built; never run	Section 2.2
Code bodies for big objects	8 builders; trial not run	Section 2.2
Blockouts, environment plan, setting	8 draws, 2 environment plans; none picked	design step
Usable people	missing: no clip clean on what is drawn	Table 2
Physics and interaction in a game engine	missing: no engine adapter; one place's physics trial, experimental	Section 5
Creator edits that survive a rebuild	missing: the edit layer is a slot	Section 2.2
Environments 2 to 4	not started	

Agent tools. With them an agent used about half the output and cost on two of three places; on the third it left more faults; one run each (Appendix E).

5 Limitations, future work and conclusion

One creator judges, and he has not yet accepted a run of environment 1. Closed models (Nano Banana Pro, Gemini 3.1 Pro) draw the concepts and close-ups and read and plan every place; they may change or go. Much of the first run was lost with its disk, and the new design step, the finish route, the shapes-first loop and the code bodies have not yet carried a place through a batch; people are not usable, light is live, objects are not interactive, and no game engine opens the result yet. Next: the remake's batch, the shapes-first loop on the ship, people, an engine adapter with physics and interaction, an edit layer the assembly carries, three more environments, open models, LEGO-Bench [15]. In short, SCORE keeps, keys and judges every stage's output, so a fault is fixed where it happens; an accepted environment will show if it holds.

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Appendix

A The hub in six rounds

The surface library, one model per real-world object and the straightness check came out of six rounds of work on the hub on 6 and 7 October 2026, each answering the creator's verdict on the round before (Table A1). In the first two rounds, the creator chose between the old and the new method side by side from the same cameras.

Table A1. The hub's six rounds. Cost is given as rented GPU time in euros and picture-model calls in dollars. Frame time is the mean frame time on an RTX 5080, in the night scene, at 1600×900.

Round	Change	Creator's verdict	Cost	Frame time
1, two walls, A/B	surface library, routing by shape, room checks	chose the new method ("Y looks much cleaner")	€0.88	7.81 ms vs 7.53
2, two walls, A/B	71-variant library, shared texture sets, roof check	chose the new method again	€1.70	8.05 ms vs 8.24
3, whole room	all stages end to end, 74 models	old models were still loaded; labels and floor fittings were wrong	€1.67	6.78 ms (old 9.22)
4, whole room	detail through the pipeline, picture colours kept	"pretty much all of this became messy"	€8.50, $5.09	9.6 ms
5, eight pieces, A/B	shape only from the pipeline, library surfaces, parts check	new method preferred for the door, notice board and porthole; three pieces still wrong	€0.21	not run
6, whole room	round 5 everywhere, one model per object, straightness check	"this looks amazing, I have no negative feedback"	€3.48, $1.07	5.6 ms

Three findings shaped the method. First, deterministic checks catch what agents miss: in round one, the share of rays leaking out through walls and roof fell from 1.53 percent of 5,817 to zero, and the colour noise within surfaces from a median of 31.1 to 3.2 percent, but only walking through the loaded room showed 26 old models still in it, which became the made-only check. Second, generated pieces that kept their pictures' colours made the room inconsistent, so surfaces come only from the library. Third, generated composite objects melted their parts together, and generated boxes leaned (one locker by 13 degrees), which led to one model per real-world object and to the straightness check. By round six the room used 0.45 million triangles, against 1.96 million in round four, and 187 MB on disk, against 487 MB.

B Check details

- **Room checks** cast rays to find light leaking out through walls and roof, find pieces in front of a real opening and roof and wall fittings more than 3 cm off their surface or turned more than 5 degrees from it, and count pieces, triangles and lamps against the room's budget.
- **Door check:** every door must stand in a wall or partition that fully separates its two sides, tested by walking a grid from one side to the other.
- **Model check:** every model must have walls at least 3 mm thick. A generated box fails if one of its sides tilts more than 2 degrees or is warped. Making a generated model solid stops if its bounding box grows more than 3 percent.

- **Likeness gate:** a made model's outline from eight sides must overlap its close-up's by at least 0.50, a least set on the hub's twelve accepted generated models with a margin of 0.05, and its DINOv2 [16] similarity must reach a least set the same way; without that calibration the gate answers unknown.
- **Finish maps (the first run's route):** a close-up's map passes when its outline overlaps the close-up's by at least 0.85 and it uses every listed finish; a part whose second finish holds 20 percent of its voted area mixes finishes and fails.
- **Bake check:** each baked part within ΔE 20 (CIE76) of its surface's colour, worn and dirtied as far as its place's wear and dirt, and within ΔE 32 of its close-up's; lightness order fails when the close-up shows one surface more than 10 L* lighter and the bake makes it more than 3 L* darker; one look fails when no two near-neutral surfaces bake far enough apart for a viewer to tell at the place's view distance, while the close-up shows them more than ΔE 24 apart. What a viewer tells is measured, not set: Stone, Szafir and Setlur measured the CIELAB difference half of their viewers see between two colours s degrees wide, per axis ND(50, s) = C + K/s, with (C, K) = (5.08, 0.75) for L*, (5.34, 1.54) for a* and (5.35, 2.87) for b* on sizes 1/3 to 6 degrees [38]; two surfaces are told apart when their difference, each axis over its ND, is 1 or more. A surface's size is its model's middle size times the root of its area share, seen from the place's nearest view. This replaced a fixed ΔE 12, which failed the wreck's access cover on a 9.3 L* step between its steel and its scorched white that its owner sees (11.3 degrees from the nearest view; half of viewers see 5.2 L* at that size); a quadrant of a standing model's base more than 3 cm off the floor fails. Glass, screens, lamps and print are judged by their own colour and left out of the order.
- **Settling and resting check:** an object that would turn more than 15 degrees or drift more than 30 cm when dropped keeps its laid pose and is marked as not resting; small debris, with no side over 1.4 m, may turn freely and drift up to 1 m. Hung objects are exempt by their inventory anchor, and fixed objects are not judged for tipping. The triage fails a foot off its support and an up axis leaning more than 3 degrees.
- **Light:** the place's median brightness within 1.5 times its concept's and every view within 2.5 times; a lamp is blown when over 20 percent of the pixels where it gives at least half the light are white, and unlit when it gives less than 0.001 percent of every view's light.
- **Motion:** no moving thing within 3 cm of a still thing or another moving thing, read every 0.25 s; rounds join within 1 mm; speeds within each kind's cited range.
- **Sound:** CLAP score against the brief at least 0.30, the tenth percentile of 55 of the environment's picks; a take no more than 0.5 s longer than its brief.
- **People:** the shell gate passes a look when 99 percent of its soft points lie within 5 cm of the fitted body; the worst point and its joint are recorded. A clip is usable when no frame tucks more than 5 mm, no hard part stretches past 1.2 times its bind length, cloth crosses cloth no more than in the earlier body's same clip, and no part's edges pass through cloth deeper than 3 mm more often than at the bind. A far-only look is judged on its joins, likeness and the far body's triangles (at most 1,220); the near outfit's triangles and shell offset are recorded, not gating. The A-pose outline is judged against the finished Pixal3D take the build starts from; the take's own loss against the raw model is recorded apart. A crowd copy stands only where it fills under a tenth of every view's height.
- **Worse than accepted:** a measure counts as worse only when it moves the wrong way by at least its tolerance, the smallest change a viewer could see: one more fault for a count; 0.02 of full scale for lightness on 0 to 1 (a halo, fill, beam or haze wash), and 5 of 255 levels for a colour gap, the step a viewer notices between two pictures; 0.05 for a brightness ratio; one percent of the picture for a share of it; a twentieth of the Earth's lit disc; 5 mm for a length.

A score no viewer reads directly, such as a likeness score, is worse by any move. Tolerances decide only whether an output waits for the creator; a check that fails still fails.

- **The creator's accept carried forward:** when the creator accepts a worse output, that accept records what was allowed, each measure's value before and after. A later output that passes every gate and is no worse than the output taken, within the tolerances, is accepted by the runner with that accept named; a move beyond it still waits for the creator.
- **What the creator acts on:** the creator rejects, accepts or waives only an output already seen on its review page. This is the creator's rule; the commands do not yet check that the page was opened.
- **People seen by the cameras (experimental):** a clip's crossings of body through cloth are counted only where a camera that will show the person can see them: the place's views at the person's cast spot, and a close front and three-quarter view. Crossings no camera sees are recorded, not counted against the clip. So far this is a measure on the stored bodies (Section 3.5); the per-clip gate does not use it yet.
- **Completion gate:** the requirements given in Section 3.6, each with its result tied to a fingerprint of the stage's files and inputs; a result made on another stage counts as unknown. The placeholder check judges every mesh a code builder draws, and only the shell builders pass unnamed.

People results in detail (experimental, Section 3.5). The body gates are joins, outline, likeness, triangles and shell. Nev's and Sefa's boot tuck bands lie 9 to 11 mm outside the trouser leg at the bind pose, in every clip before any motion. Scored on drawn surfaces only, no clip of any look is usable: skin seams open and settled cloth crosses itself beyond the bind. Bound as one closed mesh, the player's limbs sink into chest and hips, 87 to 563 crossing points visible from the places' cameras and close views per clip, the armpit surface stretching between arm and chest. Where hair or cloth meets the head or hand skin the two sides are weighted differently and open as the joints move (cloth seams never open); openings over 10 cm on three looks are not yet confirmed by a render. Binding the take as one closed mesh removes the openings. Pushing points out of the look's own volume, limiting the offset from SOMA-X at the armpits and inner thighs, fold-centred weights and dual-quaternion skinning each left the visible crossings the same or higher. A close-up drawn in a wider A-pose (arms 45 degrees below level, feet 35 cm apart) gives a take with fewer crossings at the bind (the player's 678 against 772) but more in motion, as the clips bring the arms further down: 10,610 visible crossings against 4,078 over the player's 19 clips, and no clip clean in either pose. The clips' joint angles stay within the normal human ranges of the American Academy of Orthopaedic Surgeons, except hip extension of 24 and 27 degrees (limit 20) in two clips.

Decisions recorded in these checks. Three people's gates are decisions of the captain, the agent session that then ran environment 1, taken on 12 and 13 October 2026 under the creator's rules and open to the creator's reversal. (1) The shell gate counts 99 percent of the soft points rather than every point, with the worst point recorded: the points beyond it were loose cloth edges (cuffs, hems, caps), which no part label names, so they could not be given a joint of their own by label. (2) A look seen only far away is judged on what is drawn far away, not on the near outfit it was made from. (3) The A-pose outline is judged against the finished take, not the raw model, so the take's own loss is not charged to the build. Earlier, the full-body cut (which looks get bodies in full and which only far ones) was set by the people's share of the budget.

C Models and licences

Every model in the framework must allow commercial use of its output (Table C1); we read each licence for its clauses on commercial use. Candidates were dropped under this rule, among them NVIDIA's Lyra 2.0 (weights for internal research only), Hunyuan3D 2.1 (licence does not apply in the European Union) and GarmentCode's fork of NVIDIA Warp (non-commercial research only), which Newton's cloth solver on upstream Warp replaced; Blender's cloth simulation [39] is kept as a second route. The one open doubt is the face LoRA, whose training data is not disclosed.

Table C1. Models in the framework.

Model	Role	Licence
Nano Banana Pro [30]	concepts, close-ups on the prop route, ground skins; in the first run, finish maps (the close-up drawn again with one flat colour per finish)	paid API
Gemini 3.1 Pro [29]	the environment concept's interview; scene inventories and plans; each model's finishes listed and labelled, matched to library surfaces, and rechecked	paid API
Qwen-Image-Edit-2511 [40]	close-ups, first, in the first run's close-up stage	Apache-2.0
Qwen3.8-27B [41]	the first run's close-up shape check judge; each part's material	Apache-2.0
Pixal3D [17]	picture to 3D	MIT; DINOv3 licence [42] for its encoder
PartCrafter [20]	parts	MIT; its non-commercial background remover not used
SegviGen [33]	laid the close-up's finish map onto the 3D shape (the first run's surface stage)	MIT, on TRELLIS.2 (MIT); DINOv3 licence for its encoder; its non-commercial background remover and nvdiffrast not used
GeoSAM2 [43]	parts of our own model, seeded by the close-up's regions (trial)	Apache-2.0
Depth Anything V2 Small [44]	ground relief	Apache-2.0
ProcFunc, Infinigen shaders	surfaces	BSD-3-Clause
MOSS-SoundEffect v2.0, CLAP	sound	Apache-2.0
World Labs [45]	optional walkable reference	paid; outputs owned by paid users
FLUX.2 klein 4B [36]	earlier prop pictures; the characters' A-pose pictures from words and their hair's clay pictures	Apache-2.0
MoGe-2 [19], SAM 3 [18]	measuring and finding objects in pictures; boxing each inventory row on its concept; counting people and crops	MIT; SAM Licence
Kimodo [37]	the characters' motion from sentences	Apache-2.0 code; NVIDIA Open Model License for the Kimodo-SOMA-RP-v1.1 weights, commercial use allowed
LLM2Vec [46] on Meta-Llama-3-8B-Instruct [47]	Kimodo's text encoder, used only while clips are made; nothing from it ships	Llama 3 Community License: commercial use allowed, the 700 million monthly users clause does not apply, "Built with Meta Llama 3" shown, no exclusion in the European Union for the text-only 8B model

Model	Role	Licence
SAM 3D Body [48], SAM3DBody-cpp [49]	body shape from a picture	SAM Licence for the weights, SAM3DBody-cpp's converted ONNX and GGUF weights included whatever their tag says; MIT for SAM3DBody-cpp's code
SOMA-X and MHR	the characters' bodies	Apache-2.0
GarmentCode [50], Newton [51] on Warp [52], Blender's cloth [39]	the clothes' sewing patterns and their drape	MIT; Newton and Warp Apache-2.0; Blender is GPL and its output is ours; GarmentCode's Warp fork (non-commercial research only) not used
Hi3DGen [53]	the characters' hair	MIT code; MIT and Apache-2.0 weights
FLUX.2 klein base 4B [54] with the refcontrol reference-depth LoRA [55]	the characters' faces	Apache-2.0, both; the LoRA's training data is not disclosed
UniRig [56]	the skeleton and skin of a quadruped	MIT code and weights
Robust Skin Weights Transfer [57]	the SOMA-X skeleton's skin weights moved onto a Pixal3D person (a trial)	MIT reference code, ported with trimesh and SciPy; libigl, whose bindings carry GPL-3.0, not used
MakeHuman eyes	the characters' eyes	CC0

The close-ups' shape check was tuned on 40 close-ups that four open picture models drew of the lab's ten objects, each scored by hand, and on Nano Banana Pro's ten pictures of the same objects. With the majority of three sampled answers, the check agreed with the hand score on 29 of the 40, passed all ten of the closed model's pictures and two marginal ones, passed none scored wrong or failed, and failed nine that were good or usable. A single answer at temperature zero was rejected: asked 51 questions twice in one batch, it changed 7 verdicts. On the lab, 7 of the closed model's first pictures drew the room or dimension lines; all 7 passed when redrawn from the crop alone.

Table C2. The close-up stage on environment 1's first run; the runner's close-up row now draws one Nano Banana Pro picture an object. Cloud is rented card time in euros; the closed model's pictures are in dollars.

Run	Close-ups	Open model accepted	Closed model accepted	Left for the creator	Cloud	Closed model
Lab, 22 rows (earlier check: one answer)	22	8	14	0	€4.12	$2.81
Environment 1's remaining rows (three places)	17	0	10	7	€3.06	$4.02

The inventory readings, the plan writings, the picture-to-3D step, the finish route, the surface bakes, the sound generation, all checks and the assembly run automatically. Coding agents (Claude Code [58]) write the framework's code and every code builder, and wrote environment 1's first-run inventories and plans; in the hub's final round, 86 of its 98 models were built in code. Some steps were still done by hand: some generated objects were turned to face the right way by eye, and a few labels were placed by hand before a rule took over that job. The concept picks, the corrections drawn on concepts and plans, and the verdicts are the creator's own choices.

D The surface library

The surface library holds families of variants, among them painted, steel, aluminium, rubber, plastic, cable, fabric, composite, glass, screen, light, print, wood, ground, foil, water, skin and hair. After the hub's second round it held 71 variants made from 14 ProcFunc recipes; it now holds 369 variants in 26 families made from 22 recipes. Each variant is a recipe with settings, takes its colours only from the palette, and is baked into base-colour, metal-roughness and normal maps at a texture density that depends on how close a viewer can get. Wear has three levels and comes from causes: edges, and kicks within 32 cm of the floor that a piece stands on. A surface the finish-map route proposes enters an append-only registry and joins the library only when the creator accepts it.

E With and without the agent tools

The tools that check an agent's work were tested on three places of environment 1: a room (the workshop, 814 objects), an outdoor place (the street, 1,198 objects) and a dense room (the hangar, 1,657 objects). They are the completion gate (Section 3.6); a triage of the resting check, which judges each object by what holds it up, read from its inventory row, and keeps only real faults [59], [60]; a pick lock, which refuses an export that would show anything but the model the creator picked for a row; and a render-and-compare, which lists the objects least like their close-ups by DINOv2 similarity [16]. In both arms, the same agent (Claude Opus 5.5 in Claude Code, headless) started from the same exported stage and inventory and was asked to make the place as complete as it could in about 50 minutes, through its layout data only; one arm had the framework from before these tools. Each arm ran once, so Table E1 shows what happened, not an average.

Table E1. One session per arm and place. Faults are the resting triage's real faults left on the final stage, measured against the arm's own data; placeholders by the first placeholder check; rows are inventory rows with no object and no written reason. Agent cost is the session's model cost; cloud is its rented machines' cost.

Place, arm	Faults, start → end	Placeholders	Rows missing	Min.	Agent cost	Output tokens	Cloud	Exports	Unknown gate calls
Workshop, without	18 → 13	0 → 0	1 → 0	36	$2.26	30.4k	€0.85	7	not used
Workshop, with	18 → 0	0 → 0	1 → 0	26	$1.42	14.3k	€0.85	2	2 of 4
Street, without	11 → 0	0 → 0	0 → 0	38	$2.40	30.0k	€0.32	9	not used
Street, with	12 → 0	0 → 0	0 → 0	39	$1.21	10.4k	€0.76	1	0 of 5
Hangar, without	14 → 4	5 → 0	0 → 0	47	$2.01	26.6k	€0.00	7	not used
Hangar, with	14 → 9	5 → 0	0 → 0	32	$2.34	29.1k	€0.64	5	3 of 6

The tools made building cheaper more often than they made the scene better. On the workshop and the street, the arm with the tools used about half the agent's output and cost, and exported the stage once or twice instead of seven to nine times, because the triage named the few real faults among hundreds of alarms. On the workshop, 13 of its 18 faults were cleared by writing into the inventory what holds a piece up, which the creator still has to confirm. On the hangar the arm without the tools left fewer faults: the arm with them spent its time replacing five placeholder slabs and trying the drop test, which failed on a room with no ground mesh (fixed after the run), and the nine faults it left are models with stray parts below their feet. No agent claimed a place was complete, so the gate's hook never had to refuse a claim. No row carries an owner's pick, so the pick lock was not tested.